

Code for Measuring Distance with HC-SR04:

```
#define TRIG_PIN D1 // Trigger pin
#define ECHO_PIN D2 // Echo pin

void setup() {
  Serial.begin(115200); // Initialize serial communication
  pinMode(TRIG_PIN, OUTPUT); // Set the trig pin as an output
  pinMode(ECHO_PIN, INPUT); // Set the echo pin as an input
}

void loop() {
  long duration, distance;
  // Clear the trigPin
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  // Send a 10us HIGH pulse to trigger the ultrasonic sensor
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  // Read the time it takes for the echo to return
  duration = pulseIn(ECHO_PIN, HIGH);

  // Calculate distance in centimeters
  distance = (duration * 0.0343) / 2; // Speed of sound is ~343 m/s or 0.0343 cm/us

  // Print the distance to the Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
```

```
Serial.println(" cm");

// Wait 500 milliseconds before the next reading
delay(500);
}
```

Code for Measuring Humidity and Temperature with DHT11/DHT22:

```
#include "DHT.h"

#define DHTPIN D4 // Pin where the data pin of DHT sensor is connected
#define DHTTYPE DHT11 // DHT11 or DHT22 (replace with DHT22 if you're using that)

DHT dht(DHTPIN, DHTTYPE); // Initialize DHT sensor

void setup() {
  Serial.begin(115200); // Start serial communication
  dht.begin(); // Initialize the sensor
}

void loop() {
  // Wait a few seconds between measurements
  delay(2000);

  // Read humidity
  float humidity = dht.readHumidity();

  // Read temperature as Celsius
  float temperature = dht.readTemperature();

  // Check if any reads failed and exit early
```

```
if (isnan(humidity) || isnan(temperature)) {  
    Serial.println("Failed to read from DHT sensor!");  
    return;  
}  
  
// Print the humidity and temperature values to Serial Monitor  
Serial.print("Humidity: ");  
Serial.print(humidity);  
Serial.print(" %\t");  
Serial.print("Temperature: ");  
Serial.print(temperature);  
Serial.println(" *C");  
}
```

Buzzer Triggers if Distance is Less Than a Threshold:

```
#define TRIG_PIN D1  
#define ECHO_PIN D2  
#define BUZZER_PIN D3  
  
void setup() {  
    pinMode(TRIG_PIN, OUTPUT);  
    pinMode(ECHO_PIN, INPUT);  
    pinMode(BUZZER_PIN, OUTPUT);  
    Serial.begin(115200);  
}  
  
void loop() {  
    long duration, distance;
```

```

// Clear the trigPin
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);

// Trigger the sensor
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

// Read the echo time
duration = pulseIn(ECHO_PIN, HIGH);
distance = (duration * 0.0343) / 2;

// Check if the distance is below a certain threshold
if (distance < 10) { // Trigger buzzer if object is closer than 10 cm
    digitalWrite(BUZZER_PIN, HIGH); // Turn on the buzzer
} else {
    digitalWrite(BUZZER_PIN, LOW); // Turn off the buzzer
}

Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

delay(500);
}

```

Code for Traffic Light:

```

#define RED_PIN D5
#define YELLOW_PIN D6

```

```
#define GREEN_PIN D7
```

```
void setup() {
```

```
    // Initialize the LED pins as output
```

```
    pinMode(RED_PIN, OUTPUT);
```

```
    pinMode(YELLOW_PIN, OUTPUT);
```

```
    pinMode(GREEN_PIN, OUTPUT);
```

```
    // Start with the red light on
```

```
    digitalWrite(RED_PIN, HIGH);
```

```
    digitalWrite(YELLOW_PIN, LOW);
```

```
    digitalWrite(GREEN_PIN, LOW);
```

```
}
```

```
void loop() {
```

```
    // Red light on for 5 seconds
```

```
    digitalWrite(RED_PIN, HIGH);
```

```
    digitalWrite(YELLOW_PIN, LOW);
```

```
    digitalWrite(GREEN_PIN, LOW);
```

```
    delay(5000);
```

```
    // Yellow light on for 2 seconds
```

```
    digitalWrite(RED_PIN, LOW);
```

```
    digitalWrite(YELLOW_PIN, HIGH);
```

```
    digitalWrite(GREEN_PIN, LOW);
```

```
    delay(2000);
```

```
    // Green light on for 5 seconds
```

```
    digitalWrite(RED_PIN, LOW);
```

```
    digitalWrite(YELLOW_PIN, LOW);
```

```
digitalWrite(GREEN_PIN, HIGH);  
delay(5000);  
}
```